

Judge0 installation!

Install Homebrew (if not already installed)

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Install Docker using Homebrew

```
brew install --cask docker
```

Launch Docker Desktop

```
open /Applications/Docker.app
```

Verify Docker installation

```
docker --version
```

Verify Docker Compose installation

```
docker-compose --version
```

Heading 1

```
# 🚀 Setting Up WSL, Docker, and Judge0 on Windows

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## **Step 1: Install WSL and Ubuntu**

1. **Enable WSL**
   - Open **PowerShell** as **Administrator** and run:
     ```powershell
 wsl --install
     ```
   - This will install **WSL** (Windows Subsystem for Linux) along with the default **Ubuntu**.

2. **Restart Your Computer**
   - After the installation, you'll be prompted to **restart your computer**.
   - Restart to complete the WSL setup.

3. **Complete Ubuntu Setup**
   - Open the **Ubuntu terminal** from the Start menu.
   - Follow the prompts to **create a UNIX username and password**.

4. **Update Ubuntu**
   - Once Ubuntu is set up, run the following commands to update all packages:
     ```bash
 sudo apt update && sudo apt upgrade -y
     ```

5. *(Optional)* **Adjust cgroup settings for better Docker compatibility**
   - Open the GRUB config file:
     ```bash
 sudo nano /etc/default/grub
     ```
   - Find the line starting with GRUB_CMDLINE_LINUX and change it to:
     ```bash
 GRUB_CMDLINE_LINUX="systemd.unified_cgroup_hierarchy=0"
     ```
   - Save and exit (Ctrl + O, Enter, Ctrl + X).
   - Update GRUB and reboot:
     ```bash
 sudo update-grub
 sudo reboot
     ```

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## **Step 2: Install Docker and Docker Compose**

1. **Install Docker**
   - Open the **Ubuntu terminal** and run:
     ```bash
 sudo apt update && sudo apt install -y docker.io
     ```
   - (Note: It's better to install docker.io package for Ubuntu.)
```

```
2. **Install Docker Compose**
- Still in the Ubuntu terminal, install Docker Compose:
  ```bash
 sudo apt install -y docker-compose
  ```

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## **Step 3: Install and Set Up Judge0**

1. **Download and Extract Judge0**
- Download the Judge0 release archive:
  ```bash
 wget https://github.com/judge0/judge0/releases/download/v1.13.1/judge0-v1.13.1.zip
  ```
- Unzip the downloaded archive:
  ```bash
 unzip judge0-v1.13.1.zip
  ```

2. **Set Up Secure Passwords**
- **Generate random passwords** for Redis and Postgres:
  - Visit [Random Password Generator](https://www.random.org/passwords/?num=1&len=32&
  - Open the `judge0.conf` file:
    ```bash
 nano judge0-v1.13.1/.env
    ```
  - Update the `REDIS_PASSWORD` with the generated password.
  - Repeat the process for `POSTGRES_PASSWORD` using a new random password.

3. **Start Judge0 Services**
- Navigate to the Judge0 folder:
  ```bash
 cd judge0-v1.13.1
  ```
- Start the database and Redis services:
  ```bash
 docker-compose up -d db redis
  ```
- Wait for a few seconds:
  ```bash
 sleep 10s
  ```
- Start the remaining services:
  ```bash
 docker-compose up -d
  ```
- Wait a few more seconds:
  ```bash
 sleep 5s
  ```

4. **Verify the Installation**
- Open your browser and visit:
  ```
 http://localhost:2358/docs
  ```
- You should see the Judge0 API documentation page, meaning your Judge0 instance is r
```